Active learning expedites cross-situational word learning–even under Zipfian distributions

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Abstract

Mathematical analyses of vocabulary growth have asked how long a learner who encounters words at random would take to learn a vocabulary, and have concluded that the time is plausible when every word is equally likely to be encountered. We show that this conclusion depends on that assumption. When a learner must wait for each word to occur, the rarest word sets the pace, and under the Zipfian word frequencies of child-directed speech the required number of learning episodes grows by a factor of about $\ln W$ for a vocabulary of W words–7.5 to 11.6 times for vocabularies of 1,000 to 60,000 words, and more for real corpora. We then ask how much of this cost a learner (or caregiver) who influences which words and referents co-occur can recover. Choosing among scenes so that the rarest word becomes as likely as possible speeds learning 6.7-fold on a dataset of captioned photographs and 235-fold on a database of labeled scenes, and this choice can be learned online. Incremental learners that need several exposures per word, and preferentially target words they do not yet know, learn about 2–10 times faster than passive learners by the time they know 50–90% of their vocabulary, and up to 19–28 times faster at 99%. When tentative guesses can be forgotten, this advantage grows, because returning to unknown words also rehearses them. How children and caregivers structure learning situations, not exposure alone, may therefore be central to how a Zipfian vocabulary is learned.

Keywords: Word learning; Cross-situational learning; Active learning; Zipf's law; Coupon collector problem

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Introduction

Children learn words at a remarkable pace. Vocabulary estimates rise from a handful of words at 12 months to several hundred at two years, roughly a thousand at three, and at least ten thousand comprehended words by five (Table 1). A commonly cited adult target is 60,000 words by age 18 (Bloom, 2000), although more recent estimates are closer to 40,000 word families (Brysbaert, Stevens, Mandera, & Keuleers, 2016). Each new word must be mapped to its meaning in situations where a word could refer to many possible things (cf. Blythe, Smith, & Smith, 2016).

Our models describe a child learning from spoken, caregiver-mediated situations. That process is most relevant before children learn to read; from school age on, reading and explicit instruction bring different input distributions and learning mechanisms. We therefore take the pre-literacy vocabulary, about 10,000 words by age five, as our primary target, and report results for 1,000 to 60,000 words, the last for comparison with earlier analyses (Blythe, Smith, & Smith, 2010; Vogt, 2012). As we show, our main conclusions concern ratios of learning times, which change only slowly with vocabulary size, so they do not depend on which target is chosen.

Sampling models of word learning

Children can use many cues to resolve what a word refers to: social cues (Hirsh-Pasek, Golinkoff, & Hollich, 2000), a bias to map novel words to whole objects (Macnamara, 1972), mutual exclusivity (Merriman & Bowman, 1989), and, in the lab, the ability to learn from a single informative instance (*fast mapping*; Carey & Bartlett, 1978; Spiegel & Halberda, 2011). These cues are thought to support *cross-situational learning*: accumulating evidence about a word's meaning across situations, as infants and adults do in the lab (L. Smith & Yu, 2008; Yu & Smith, 2007). Models of this process differ in whether they test one hypothesis per word (e.g., Trueswell, Medina, Hafri, & Gleitman, 2013) or accumulate graded associations (e.g., Kachergis, Yu, &

Shiffrin, 2012; McMurray, Horst, & Samuelson, 2012), and the two families often mimic each other (Yu & Smith, 2012).

A complementary line of work abstracts away from these details to ask how long learning an adult-sized vocabulary should take (Blythe et al., 2016, 2010; Vogt, 2012). Blythe et al. (2010) analyzed a learner who maps each word correctly on its first encounter, which gives the fastest possible learning rate. If every word is equally frequent, such a learner needs about 940,000 episodes to learn 60,000 words, about 142 per day for 18 years. They also noted that a Zipfian word distribution should be slower by a factor of 11.58 for this vocabulary size. Vogt (2012) studied a learner who narrows a word's meaning by eliminating candidate referents across episodes, with both words and candidate referents drawn from Zipfian distributions, and found that learning time grows super-exponentially with the ratio of context size to the number of possible meanings. Vogt's slowdown comes from frequent distractors that keep co-occurring with a word's target; the slowdown we analyze first is different, arising from rare words under fast mapping, where distractors play no role. Real input is skewed in both respects: child-directed speech is Zipfian across languages (Lavi-Rotbain & Arnon, 2023), and so are the objects infants see (Clerkin, Hart, Rehg, Yu, & Smith, 2017).

Active learning

These sampling models treat the learner as a passive recipient of randomly drawn input. Yet learners shape what they experience (Gottlieb & Oudeyer, 2018; Gureckis & Markant, 2012). Infants direct their gaze, reach, and requests toward objects, and in play with caregivers often bring a single object to dominate their view (Yu, Smith, Shen, Pereira, & Smith, 2009). Young children benefit from control over their own learning (Partridge, McGovern, Yung, & Kidd, 2015), as do adult word learners (Kachergis, Yu, & Shiffrin, 2013; Zettersten & Saffran, 2021), and language development may be better described in terms of a child who selects informative input than one who receives it (Foushee, Srinivasan, & Xu, 2023). In machine learning, choosing which examples to learn from is a well-studied way to reduce the data needed to learn (Settles, 2012).

The present study

We ask how a skewed word distribution affects learning time, and how much of the cost an active learner can recover. We make three contributions.

- **The cost of Zipfian input.** For a learner who learns each word on first encounter, learning time is governed by the rarest word, and a Zipfian distribution costs a factor of about $H_W \approx \ln W$ relative to a uniform one. We confirm this with child-directed speech and written-language corpora.

- **Static active control.** A learner who chooses which scenes to experience can raise the probability of the rarest word. On two real datasets this speeds learning 6.7- to 235-fold, and a learner can estimate the required statistics online.

- **Adaptive active control.** Two incremental learners that need several exposures per word learn much faster when they target words they do not yet know, on synthetic Zipfian vocabularies of 1,000 to 60,000 words and on real image captions. The advantage grows with the completion target and is larger when tentative guesses can be forgotten.

Framework

All models share one scenario. There are W words and W referents, each word paired with exactly one referent (following Blythe et al., 2010). The learner experiences a sequence of *episodes*. In each, they hear one word in a *situation* containing C referents: the word's own referent (the *target*) and $C - 1$ *distractors*.

Models differ in two ways (Figure 1). First, a *fast-mapping* learner learns a word the first time it is heard; it is an idealized bound on the fastest learning possible, not a mechanistic account of how distractors are ruled out. An *incremental* learner accumulates evidence across episodes. Second, a *passive* learner encounters situations drawn at random from fixed word and referent distributions, while an *active* learner (or caregiver) influences which situations occur. We distinguish two kinds of active control. *Static* control is a fixed preference over the kinds of scene the learner experiences, which

reshapes the overall frequency of words. *Adaptive* control depends on the learner's current state, for instance choosing to play with a toy whose name is not yet known.

As a running example, imagine a toddler learning the names of five toys: a ball, a cup, a dog, a car, and a hat. A caregiver says "ball" while the ball, cup, and dog are in view. A fast-mapping learner is simply credited with learning "ball." An incremental learner notes that "ball" co-occurred with all three toys, and needs later episodes in which "ball" occurs without the cup or the dog. Whether the child hears "ball" at all depends on which toys are nearby, and so on whether toys are picked up at random or chosen.

Passive fast-mapping under Zipfian input

Uniform word frequencies

If each episode presents one of W equally likely words, and each word is learned on its first occurrence, then learning the whole vocabulary is the coupon collector's problem (Blom, Holst, & Sandell, 1994). The expected number of episodes is

$$E[T] = W \sum_{i=1}^{W} 1/i = W H_W \approx W \ln W, \tag{1}$$

where H_W is the Wth harmonic number. For $W = 60{,}000$, $E[T] = 694{,}759$.[1]

Arbitrary word frequencies

Now let word i occur with probability p_i. The number of episodes T by which a proportion $1 - \epsilon$ of learners have learned every word is approximately the root of

$$\prod_{i=1}^{W} \left(1 - (1 - p_i)^T\right) = 1 - \epsilon \tag{2}$$

(Appendix A). Write

$$f_{W,\epsilon}(x) := \frac{\log\left(1 - (1-\epsilon)^{1/W}\right)}{\log\left(1 - x\right)},$$

[1] Blythe et al. (2010)'s figure of about 940,000 is not a mean but the number of episodes by which 99% of learners have finished; Equation 3 below with $\epsilon = 0.01$ gives 936,000, in agreement.

the number of episodes needed when all W words have probability x. For uniform frequencies,

$$T = f_{W,\epsilon}(1/W), \tag{3}$$

and $\epsilon = 1/2$ gives the median. For any other distribution, Equation 2 has no closed form, but its root is bounded by

$$f_{W,\epsilon}(\max p) \leq T \leq f_{W,\epsilon}(\min p) =: T_+. \tag{4}$$

In words, learning time is set by the rarest word: every more frequent word will usually have occurred long before it. The expected time for unequal probabilities is classical (Boneh & Hofri, 1997; Flajolet, Gardy, & Thimonier, 1992); the bound is useful here because it needs only one number, $\min p$, and so applies directly to empirical distributions. Because $\min p \leq 1/W$ for any distribution, the uniform distribution is the fastest to learn.

The cost of a Zipfian distribution

Under a Zipfian distribution with exponent a, $p_r = r^{-a}/H_{W,a}$, where $H_{W,a} = \sum_{i=1}^{W} i^{-a}$, so $\min p = W^{-a}/H_{W,a}$. For small probabilities $f_{W,\epsilon}(x) \propto 1/x$, so the cost of Zipfian relative to uniform frequencies is

$$\frac{T_+^{\text{Zipf}}}{T^{\text{uniform}}} \approx W^{a-1} H_{W,a}, \tag{5}$$

which for the standard exponent $a = 1$ is simply $H_W \approx \ln W + 0.58$. This is 7.5 for $W = 1{,}000$, 9.8 for 10,000, and 11.6 for 60,000–the factor of 11.58 reported by Blythe et al. (2010) for 60,000 words. The penalty grows only logarithmically with vocabulary size, so it is nearly the same whether the target is a toddler's or an adult's vocabulary (Table 2).

The bound is tight in this case. For $W = 60{,}000$, $a = 1$, and $\epsilon = 0.01$, $T_+ = 1.08 \times 10^7$, while the exact root of Equation 2 is 8.97×10^6. Steeper distributions are much harder: $\log T_+$ is convex in a (Appendix A), so T_+ grows faster than exponentially with the exponent (Figure 2).

Empirical word frequencies

Real distributions are more extreme than the idealized one. Table 2 applies Equation 4 to word frequencies from CHILDES, a corpus of child-directed speech (MacWhinney & Snow, 1990), and from Google Books (Michel et al., 2011) (Figure 3). For CHILDES we excluded words occurring once, which are the most likely to be specific to one household, such as a pet's name. Because T_+ depends only on the rarest word, it is sensitive to how a corpus treats rare words: pooling many families may make some words look rarer than any one child finds them, while a finite corpus may miss rare words that every child hears. Excluding all words with fewer than 11 occurrences lowers T_+ to near the idealized Zipfian value; the Google Books list, which omits every word with fewer than 100,000 occurrences, still gives a value in the same range as the CHILDES estimate. Across these choices, the empirical penalty relative to uniform frequencies is 67 to 138 times, compared with 10 to 12 for an idealized Zipf distribution of the same size. We therefore treat the corpus estimates as confirming the direction and rough size of the Zipfian penalty, not as precise bounds.

What counts as an episode

Is 10^6–10^7 episodes a lot? For the pre-literacy target ($W = 10{,}000$ over five years of 8-hour days), the Zipfian bound implies about 93 episodes per hour, against about 9 for uniform frequencies; for 60,000 words over 18 years it implies 206 per hour. Children hear from several hundred to over two thousand word tokens per hour (Hart & Risley, 1995), so if every token were an episode, these rates would be easily met. But an episode here is a situation in which a word and its referent co-occur informatively enough to learn from, and these are far rarer than tokens. Fitting a waiting-time model to vocabulary norms in 13 languages, Mollica and Piantadosi (2017) estimate that English-learning children need about 8–11 such effective learning instances per early word, arriving at only about 0.5 per word per month (see also Hidaka, 2013)–roughly 1.5 per hour across the 680 words of the English production inventory. By this measure even the uniform case is demanding. We therefore do not rest our argument on whether

any absolute number is plausible. What the analysis does establish is that skewed input multiplies the required number of informative episodes by an order of magnitude or more, and the question is what reduces that factor.

Static active control: choosing scenes

Formulation

Suppose the learner (or caregiver) can choose among S kinds of scene–for instance, playing in the kitchen, the park, or the garage–and in scene j the word-referent pair i occurs with probability p_{ij}. Collect these in the $W \times S$ matrix P. A learner who chooses scene j with probability q_j hears words with marginal distribution Pq. A passive learner, who chooses scenes at random ($q = \mathbf{1}_S/S$), hears whatever that produces. Because learning time is governed by the rarest word, the best static policy is

$$\hat{q} = \arg\max_{q} \min(Pq),$$

the mixture of scenes that makes the least likely word as likely as possible.[2] This policy does not speed the learning of any word once heard; it changes which words are heard, spreading exposure more evenly over the vocabulary. We compute $\hat{q}$ by iterated linear programming (Appendix B), and measure its benefit by the ratio

$$R = \min(P\hat{q}) / \min(P\mathbf{1}_S/S),$$

which, since $f_{W,\epsilon}(x)/f_{W,\epsilon}(y) \approx y/x$ for small probabilities, is approximately how many times faster the active learner is. Random scene choice is a generous passive baseline: because $\min(Pq)$ is concave in q, a learner whose scene preferences are themselves random does no better on average than one who chooses uniformly (Appendix B).

Real scenes

The SUN database (Xiao, Hays, Ehinger, Oliva, & Torralba, 2010) is a large collection of photographs of everyday places, each assigned a scene category (e.g.,

[2] In game-theoretic terms, $\hat{q}$ is the max-min strategy of the two-player zero-sum game with payoff matrix P (Maschler, Eilon, & Shmuel, 2013).

kitchen, bedroom, street) and annotated with the names and outlines of the objects it contains. From the per-image object annotations of 25,294 images, we built P with $W = 3{,}458$ object categories and $S = 1{,}111$ scene categories, each column giving the probability of each object in that scene category. Under random scene choice, $\min(Pq) = 8.30 \times 10^{-9}$; under $\hat{q}$, $\min(P\hat{q}) = 1.95 \times 10^{-6}$, so active scene choice is about 235 times faster. Figure 4 shows the two marginal distributions: under random choice the tail falls off exponentially, under $\hat{q}$ as a power law. The passive marginal is far more skewed than an idealized Zipf distribution over the same number of objects ($T_+ = 1.54 \times 10^9$ vs. 3.85×10^5), and even the optimal learner (6.52×10^6) does not reach the idealized case: static control recovers much, but not all, of what real-world skew costs.

A second dataset gives a smaller but consistent result. Flickr8k (Hodosh, Young, & Hockenmaier, 2013) contains 8,108 photographs, each described by five people. Treating words as referents and photographs as scenes ($W = 4{,}438$ words), $\min(Pq) = 2.68 \times 10^{-6}$ under random photo choice and $\min(P\hat{q}) = 1.80 \times 10^{-5}$ under $\hat{q}$, a 6.7-fold speedup, and the passive marginal is again harder than an idealized Zipf distribution of the same size (9.4 times). The difference between the datasets likely reflects their units: a SUN scene category pools object statistics across many images, giving $\hat{q}$ more to work with than a single photograph with five captions.

Learning the scene statistics online

$\hat{q}$ assumes the learner knows P. To see whether this matters, we simulated an *online* learner who estimates P from experience. We generated a synthetic P with $W = 1{,}000$ referents and $S = 100$ scenes, each column a randomly permuted Zipf distribution with exponent drawn from $[1, 1.5]$. The learner starts with uniform scene choice and, after every 1,000 episodes, re-estimates P from its sample counts and recomputes q from that estimate. Across 100 runs, the online learner was on average as fast as the learner who knows P, and almost always faster than the passive learner, which needed about twice as many episodes (Figure 5). This modest twofold gain,

compared with 235-fold on SUN, shows that the benefit of static control depends on the structure of P: when every scene has a similar Zipfian profile, as here, there is little room to rebalance the marginal.

Adaptive active control in incremental learners

The analyses so far assume that a word is learned the moment it co-occurs with its referent, so learning time measures only how often the right pair is sampled. We now relax this assumption, using learners that need several exposures per word, and ask whether targeting unknown words still helps. As in all models in this paper, words are learned independently of one another; we return to this in the Discussion.

Learners

The *eliminative* learner (Siskind, 1996; Vogt, 2012) keeps, for each word, the set of referents that have been present every time the word was heard, and removes any referent absent from the current situation. A word is learned once one candidate remains.

The *guess-test* learner, a propose-and-verify model (e.g., Trueswell et al., 2013), keeps only one guess per word. A word without a guess proposes a referent from the current situation that no other word currently claims. A guess is discarded if the word is later heard without it, since the true referent is always present when its word is heard. A word is learned once its guess is correct. Because words compete for unclaimed referents, this learner can make errors that later need correcting.

In each episode a target word is chosen, and $C - 1$ distractors are drawn from the referent distribution, excluding the target. Target selection is *passive* (by frequency) or *active* (uniformly among words not yet known).

Simulation design

Word and referent frequencies were Zipfian, $p_r \propto r^{-a}$, with $a \in \{0, 0.5, 1, 1.5\}$ ($a = 0$ is uniform). Most simulations used $W = 1{,}000$ words, about a three-year-old's vocabulary; the eliminative learner was also run at $W = 10{,}000$ and 60,000. Situations

contained $C = 10$ referents.[3] We ran 30 replications per cell and recorded episodes to learn 50%, 90%, and 99% of the vocabulary. Runs at $W \geq 10{,}000$ used a faster reimplementation of the eliminative learner, validated against the reference implementation on all cells both could complete.

Results

Table 3 gives episodes to learn 99% of the vocabulary. The pattern matches the fast-mapping analysis. Under passive sampling, a Zipfian vocabulary takes about 13 times as long as a uniform one at every size, and active target selection removes most of this cost: 19-fold at 1,000 words, 24-fold at 10,000, and 28-fold at 60,000. The active advantage grows with vocabulary size roughly in proportion to H_W (7.5, 9.8, 11.6), the same quantity that sets the Zipfian penalty under fast mapping (Equation 5). The reason is the same: under passive sampling, a learner waits ever longer for its rarest words, while an active learner never spends an episode on a word it already knows. Under random context each word needs about the same 3–4 exposures regardless of its frequency, so the active learner's time is close to the total number of exposures needed.

Varying the exponent (Figure 6) gives the same dose-response pattern as the fast-mapping analysis: passive learning slows steeply as a increases (eliminative: 6,676 episodes at $a = 0$, 10,795 at $a = 0.5$, 89,454 at $a = 1$), and active selection's advantage grows with it.

The guess-test learner shows the same pattern with smaller magnitudes. At $a = 1$ it is faster than the eliminative learner under passive sampling (38,037 episodes), and active selection speeds it up 4.9-fold (7,747 episodes). Because it needs only one correct guess rather than the elimination of every competitor, it also copes better with steep skew: at $a = 1.5$, active guess-test learning takes 37,570 episodes, against 296,637 for the eliminative learner.

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[3] Larger contexts change which forms of control matter: at $C = 100$ under Zipfian distractors, elimination does not finish unless the distractors themselves are controlled. We analyze this in a companion paper.

Real co-occurrence structure

To test adaptive targeting on natural input, we ran the eliminative learner on Flickr8k captions, treating words as their own referents. Each caption is a situation, and updates every content word it contains: a word's candidates are the words present in every caption containing it so far. After excluding function words and words occurring once, 4,318 content words remain, averaging 5.76 per caption, close to our $C = 10$. Of these, 3,598 (83%) can be learned by elimination at all; the rest share every caption with some other word. A passive learner samples a photo at random and reads one of its captions; an active learner targets the rarest learnable word it does not yet know, samples a photo described with that word, and reads a caption containing it. Across 30 replications, the passive learner needed 10,386, 60,719, and 151,671 episodes to learn 50%, 90%, and 99% of the learnable vocabulary, and the active learner 2,779, 5,643, and 6,528–speedups of 3.7, 10.8, and 23 times.

How much of the vocabulary matters

The 99% criterion is dominated by the last, rarest words. Table 4 therefore gives the speedup at 50% and 90% of the vocabulary as well, using the same simulations. Means and medians agree closely throughout (ratio 0.99–1.09), so the means are not driven by a few slow runs. Active selection already helps at 50% (1.7–4.1 times faster), and the advantage grows steadily toward completion. The 99% figures are therefore the upper end of the effect, not a typical value. Real children have ways to learn the last few rare words that our models lack–bootstrapping from known words, explicit correction, instruction–and these matter most exactly where frequency-driven sampling is slowest. The speedups at 50–90% are the better guide to what active selection does for a functional vocabulary.

Forgetting

All learners so far keep a tentative guess indefinitely. We added forgetting to the guess-test learner, whose single guess per word is a natural target for decay. In the

learner as described, a word counts as learned the moment its guess is correct, so forgetting could only erase wrong guesses, which disconfirmation removes anyway.[4] We therefore required a guess to *lock in* by surviving two consecutive re-exposures of its word. Until then, an unlocked guess decays back to "no guess" with a constant per-episode hazard p if its word is not heard, so it survives t episodes without rehearsal with probability $(1-p)^t$; each confirmation restarts this clock. Locked guesses do not decay. A word is learned once its guess is locked and correct. Because the learner cannot know whether a guess is correct, active selection here targets words whose guesses are not yet locked.

Active selection, by returning to words that are not yet locked, rehearses them. Passive sampling does not: under Zipfian frequencies a rare word may go thousands of episodes between exposures. At $C = 10$, $a = 1$, the rarest word's expected gap between passive exposures is about 7,485 episodes, and 97% of words have gaps longer than 200. We therefore tested half-lives $t_{1/2} \in \{20{,}000, 5{,}000, 1{,}000, 200\}$ episodes, with 30 replications per cell and a cap of two million episodes, and report medians, counting runs that never reached a checkpoint as slower than the cap.

Forgetting slowed active learning modestly and passive learning greatly (Table 5, Figure 7). Requiring confirmation slows both learners even without forgetting (compare 38,037 and 7,747 episodes without lock-in). With the shortest half-life, active learning took about twice as long as without forgetting, while passive learning slowed more than 14-fold already at $t_{1/2} = 1{,}000$, and at $t_{1/2} = 200$ no passive run reached 90% of the vocabulary. The speedup at 99% grew from 4.4 times without forgetting to 56 times at $t_{1/2} = 1{,}000$ and more than 70 times at $t_{1/2} = 200$, and widened at every checkpoint. Requiring three confirmations instead of two gave the same pattern, if anything stronger. The size of the effect follows from the ratio between the forgetting half-life and how long a tentative guess must wait to be rehearsed, which under passive sampling depends on the word's rarity and under active selection does not.

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[4] Adding forgetting to this learner left both active and passive learning unchanged at every half-life tested (passive: 37,642–38,774 episodes to 99%; active: 6,963–8,166).

In 2–4 of 30 active runs per condition with two-confirmation locking (none with three), learning stalled just short of 99%: a few words had locked in a wrong guess confirmed by chance, holding the true referents of other words, and an active learner never revisits words it believes it knows. The medians are unaffected, but the stall shows a limit of purely active selection: some passively encountered input, which real learners always have, is needed to correct confident errors.

Discussion

When a learner must wait for each word to occur, the rarest word sets the pace, and a Zipfian vocabulary costs a factor of about $H_W \approx \ln W$ over a uniform one, more for real corpora. This extends Blythe et al. (2010)'s analysis from the uniform case to any distribution, and explains their Zipfian factor of 11.58 as $H_{60{,}000}$. A learner who shapes their input can recover much of this cost, either statically, by favoring scenes that make rare words more likely, or adaptively, by targeting words not yet known. The adaptive advantage holds for two incremental learners, on synthetic vocabularies of 1,000 to 60,000 words and on real captions, and grows when tentative guesses can be forgotten.

Why the estimates range from 2- to 235-fold. The size of the benefit depends on what kind of control is available and on how much of the vocabulary must be learned. Static control can only rebalance the word distribution as far as the available scenes allow: 235-fold on SUN, whose scene categories differ greatly in their objects, 6.7-fold on individual Flickr8k photographs, and twofold on a synthetic matrix whose scenes all have similar profiles. Adaptive control does not depend on scene structure, but its benefit grows with the completion target, from 2–4 times at 50% to 5–23 times at 99%. What is stable is the direction: active control helps much more under Zipfian than under uniform input (Table 3).

What the corpus analysis shows. If children and caregivers do shape what is said, then CHILDES frequencies already reflect that shaping, and analyzing them as random samples is partly circular. This affects only static control, which changes the

overall word distribution. Adaptive control changes *when* each word is heard, targeting it while it is still unknown, and can speed learning even if the pooled distribution remains Zipfian. Partly for this reason, we rest our conclusions on ratios of learning times rather than on whether any absolute number of episodes is plausible.

Simplifying assumptions. Our models share the assumptions of this modeling tradition (Blythe et al., 2016, 2010; Vogt, 2012). First, each word has exactly one referent. Synonyms and morphologically related words reduce the effective number of independent items, but each must still be encountered, and because the Zipfian penalty grows only with $\ln W$, a smaller effective vocabulary changes our ratios little. Second, the learner must learn a particular set of W words. If it may instead learn any W of a larger set, the task is easier only if it concentrates on the most frequent words–itself a form of active selection. Third, words are learned independently. Children's knowledge of one word plausibly helps with others, and cross-word mutual exclusivity in particular can speed elimination dramatically (Reisenauer, Smith, Smith, & Blythe, 2013). When we added it to our eliminative learner at $C = 10$, it sped learning 3.9-fold for passive and 3.0-fold for active targeting (to 23,184 and 1,569 episodes) and reduced the passive Zipfian penalty from 13 to 4.8 times; active targeting remained 15 times faster. Hendrickson and Perfors (2019) found that adults learn Zipfian word sets *more* easily than uniform ones in cross-situational experiments, which they attribute to the same mechanism: frequent words, learned quickly, help disambiguate rare ones (see also Wolters, Lavi-Rotbain, & Arnon, 2024). Our simulations narrow but do not reverse the penalty, perhaps because of differences in scale (60 words vs. 1,000) or in memory. Both results suggest that interactions among words do considerable work in making Zipfian vocabularies learnable.

Do children select input in this way? Infants attend selectively, preferring stimuli of intermediate rather than minimal or maximal predictability (Kidd, Piantadosi, & Aslin, 2012), and self-directed information search adapted to the environment is documented from about age three (Ruggeri, 2022). Caregivers and children jointly create views dominated by one object (Suanda, Barnhart, Smith, & Yu,

2018; Yu et al., 2009). But the evidence is mixed. A recent, well-controlled study found no advantage of active over passive selection for infants' retention of novel word-object pairs (Bazhydai, Wong, Altmann, Jones, & Westermann, 2026), and adults choosing their own study schedules often choose poorly (Tauber, Dunlosky, Rawson, Wahlheim, & Jacoby, 2013). Our policies are idealized, choosing the least-known word on every episode among the whole vocabulary, so our estimates describe how large a benefit active control *could* confer, not how much children realize. In a companion paper we ask how much of it survives more realistic control: choosing only some of the time, among the objects at hand, with costs for switching scenes, along a developmental path from passive to active, and over what accompanies a word as well as the word itself.

Scene selection could be carried out by the child, the caregiver, or both as they jointly choose objects and places. Egocentric recordings of everyday play (L. B. Smith, Jayaraman, Clerkin, & Yu, 2018; Suanda et al., 2018; Yu et al., 2009) could be used to estimate how much such selection actually reshapes a child's input, and so to test whether children create, in effect, a curriculum for learning their vocabulary.

Data and code availability

Simulation code and results are available in the project repository (`corpusXSL/analysis`).

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Table 1

Representative vocabulary-size estimates across development. Comprehension and production are measured differently and are not directly comparable; see cited sources for instrument details.

Age	Estimate	Metric	Source
12 months	∼7 words	Production (CDI)	(Frank, Braginsky, Yurovsky, & Marchman, 2017)
18 months	50–120 words	Production (CDI)	(Frank et al., 2017)
18 months	100–150 words	Comprehension	(Hulit & Howard, 2002)
24–30 months	220–530 words	Production (CDI)	(Frank et al., 2017)
3 years	∼1,000 words	Comprehension	(Hart & Risley, 1995)
5 years	≥10,000 words	Comprehension	(Shipley & McAfee, 2015)
18 years / adult	∼60,000 words	Commonly cited target	(Bloom, 2000)

Table 2

Upper bound T_+ on episodes for 99% of fast-mapping learners to learn every word ($\epsilon = 0.01$), and its ratio to a uniform distribution over the same number of words.

Distribution	W	$\min p$	T_+	Ratio to uniform
Zipfian, $a = 1$	1,000	1.3×10^{-4}	8.6×10^{4}	7.5
Zipfian, $a = 1$	10,000	1.0×10^{-5}	1.35×10^{6}	9.8
Zipfian, $a = 1$	60,000	1.44×10^{-6}	1.08×10^{7}	11.6
CHILDES, words occurring ≥ 2 times	27,673	2.6×10^{-7}	5.67×10^{7}	138
CHILDES, words occurring ≥ 11 times	10,179	1.5×10^{-6}	9.52×10^{6}	67
Google Books, $\geq 10^5$ occurrences	97,565	1.3×10^{-7}	1.20×10^{8}	76

Table 3

Mean episodes to learn 99% of a W-word vocabulary, eliminative learner, $C = 10$, 30 replications per cell.

Condition	$W = 1{,}000$	$W = 10{,}000$	$W = 60{,}000$
Zipfian ($a = 1$), passive	89,454	830,515	5,089,443
Zipfian ($a = 1$), active	4,695	34,392	180,367
Uniform, passive	6,676	66,549	398,655
Uniform, active	2,068	19,966	119,399
Active speedup, Zipfian	19.1x	24.1x	28.2x
Active speedup, uniform	3.2x	3.3x	3.3x
Zipfian penalty, passive	13.4x	12.5x	12.8x

Table 4

Active/passive speedup (ratio of mean episodes) at three completion checkpoints. Synthetic rows: Zipfian ($a = 1$), $W = 1{,}000$, $C = 10$; median-based speedups agree to within 4%. Flickr8k row: the 3,598 learnable content words.

	Proportion of vocabulary learned		
Learner	50%	90%	99%
Eliminative	4.1x	9.2x	19.1x
Guess-test	1.7x	3.0x	4.9x
Eliminative, Flickr8k captions	3.7x	10.8x	23.2x

Table 5

Median episodes for the guess-test learner (guesses lock after two consecutive confirmations; unlocked guesses decay if unrehearsed) to learn 50%, 90%, and 99% of a Zipfian ($a = 1$), $W = 1{,}000$-word vocabulary, $C = 10$, by forgetting half-life $t_{1/2}$ (episodes). ">2M": no run reached the checkpoint within two million episodes.

	Active			Passive			Speedup
$t_{1/2}$	50%	90%	99%	50%	90%	99%	at 99%
No forgetting	7,517	10,368	12,758	14,851	34,382	55,814	4.4x
20,000	7,589	10,670	13,700	16,142	39,802	74,310	5.4x
5,000	7,776	10,012	11,303	19,803	63,236	139,282	12.3x
1,000	10,310	13,136	14,510	49,325	297,240	815,432	56.2x
200	21,982	26,269	27,059	412,969	>2M	>2M	>70x

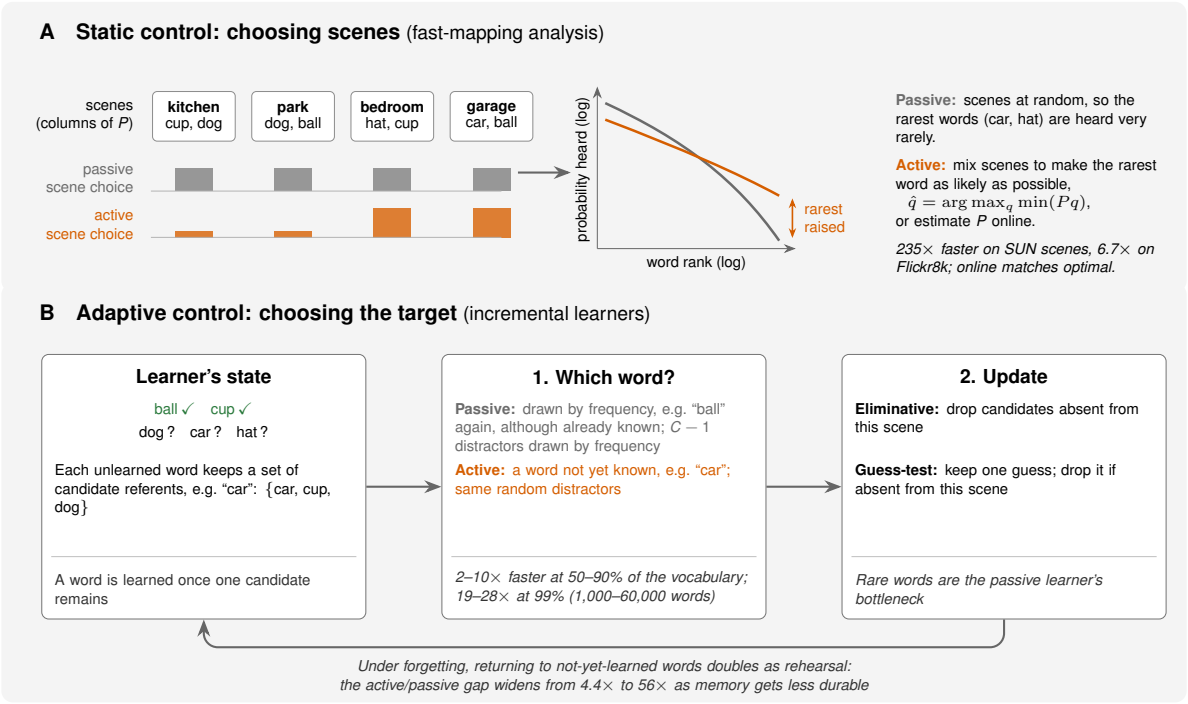


Figure 1. The two kinds of active control analyzed in this paper, illustrated with a toddler learning the names of five toys. (A) Static control: the learner or caregiver chooses among scenes, each with its own mix of objects. Choosing scenes at random leaves the rarest words very rare; the mixture $\hat{q}$ makes the rarest word as likely as possible. (B) Adaptive control in incremental learners: each episode's target word is chosen passively (by frequency) or actively (among words not yet known), and the learner then updates each word's candidate referents. Grey: passive; orange: active. Italic notes give the main result for each.

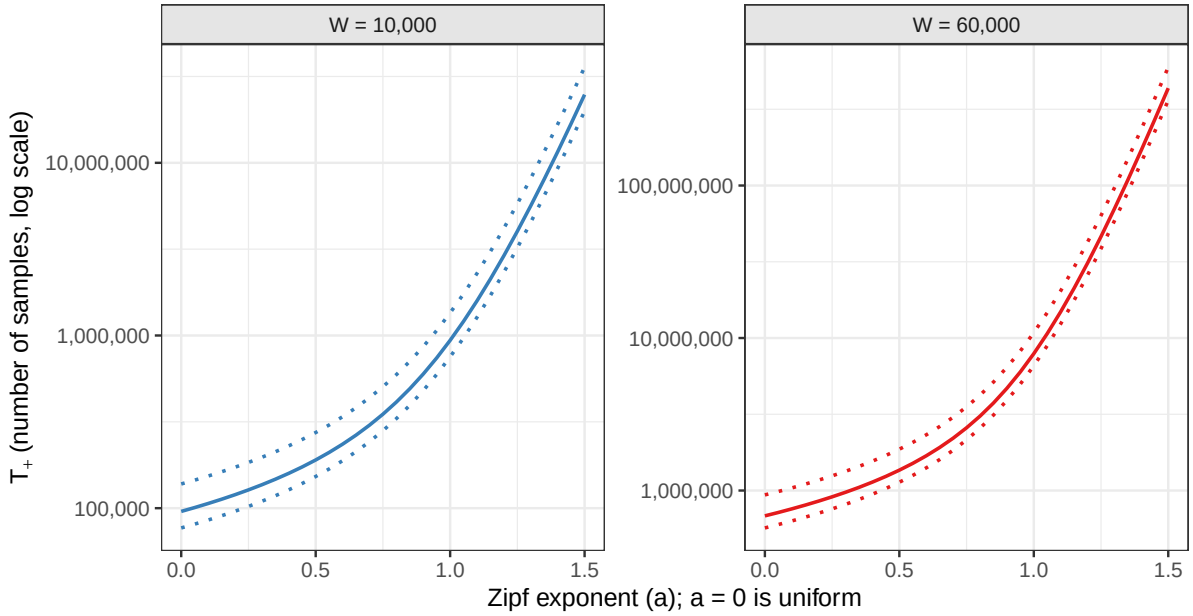


Figure 2. Episodes needed for a fast-mapping learner to learn every word of a vocabulary of $W = 10{,}000$ or 60,000 words drawn from a Zipf distribution $p_k \propto k^{-a}$, as a function of the exponent a ($a = 0$ is uniform). Solid line: median ($\epsilon = 0.5$); dotted lines: 1% and 99% of learners finished ($\epsilon = 0.99$ and 0.01). Values are the exact roots of Equation 2.

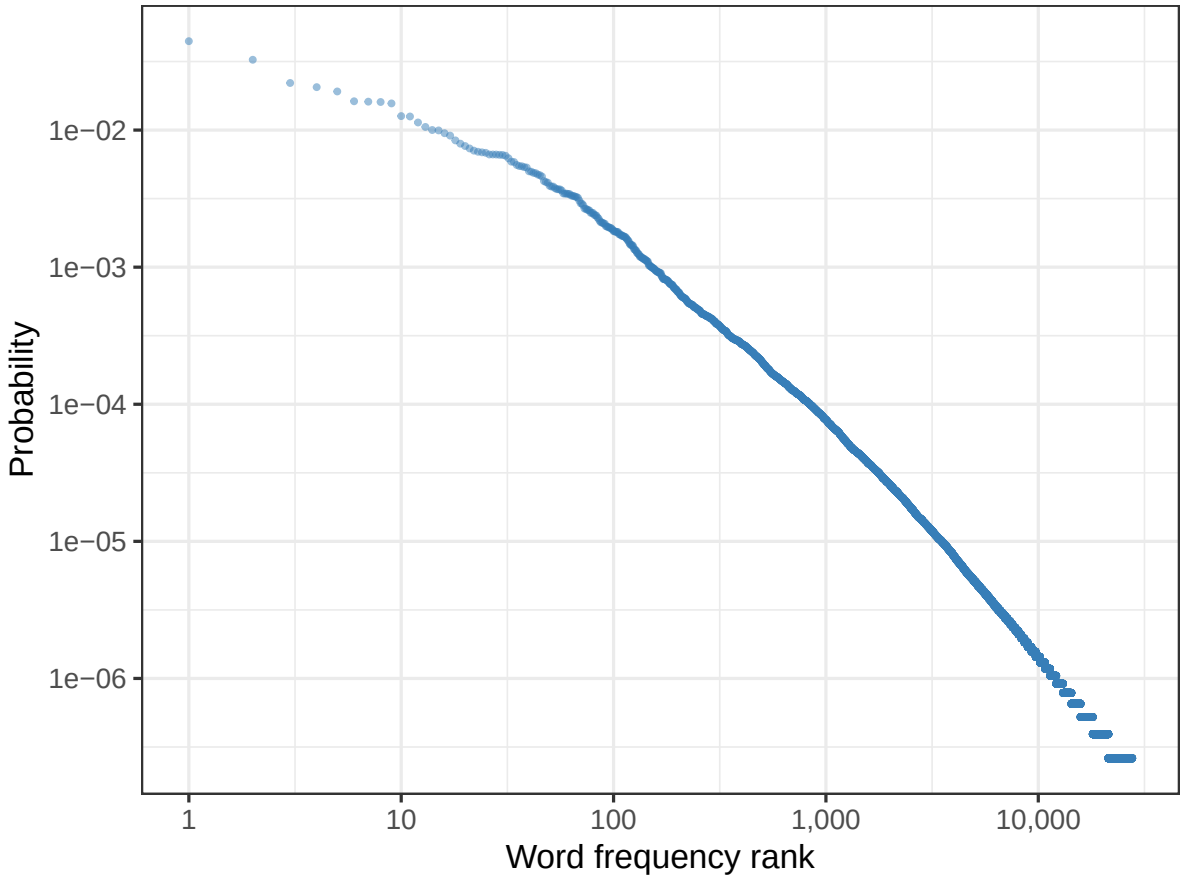


Figure 3. Word frequency (rank vs. probability, log-log) in a CHILDES child-directed speech export, excluding words that occur once (27,673 words).

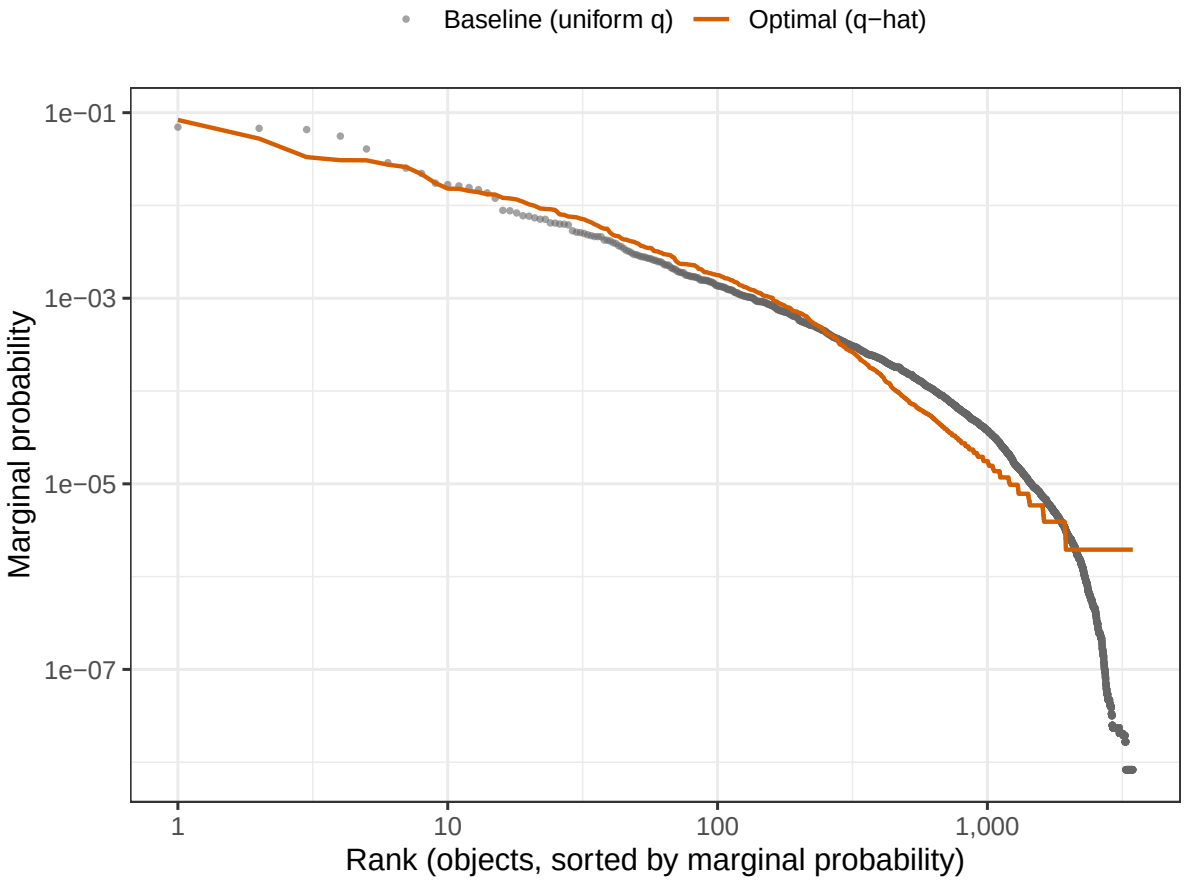


Figure 4. Marginal probability of objects in the SUN database under the optimal scene mixture $\hat{q}$ (line) and under random scene choice (dots).

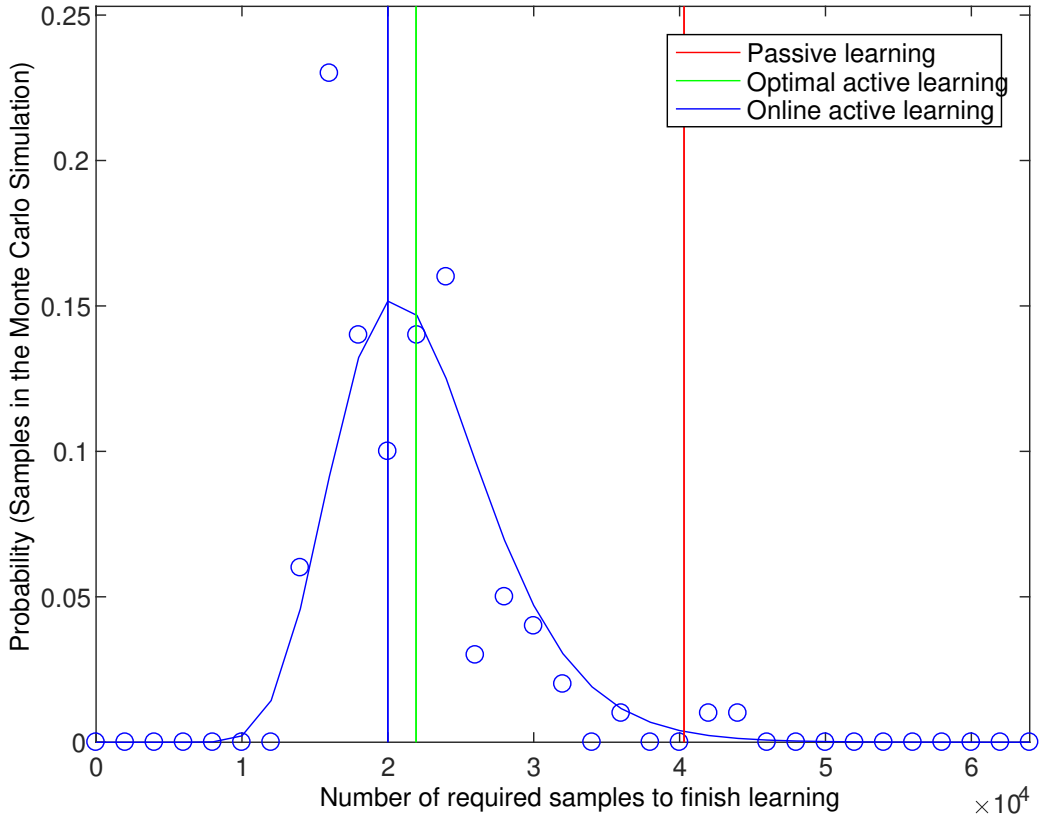


Figure 5. Distribution of the number of episodes to finish learning for passive (red), optimal (green), and online active learners (blue), $W = 1{,}000$, $S = 100$. On average, online active learning was as fast as optimal active learning.

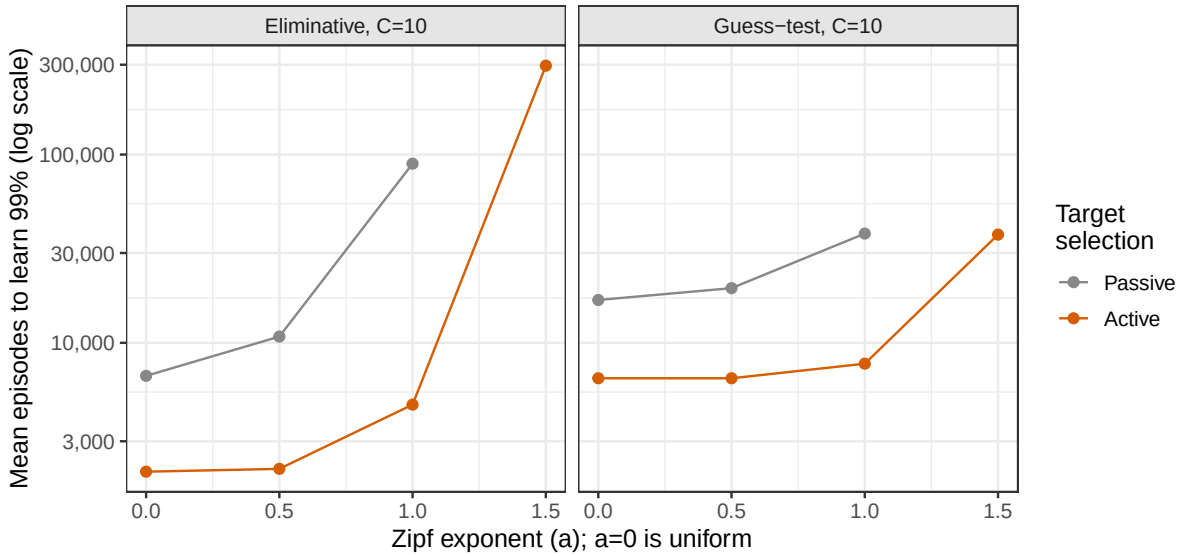


Figure 6. Mean episodes (log scale) to learn 99% of a 1,000-word vocabulary as a function of the Zipf exponent a ($a = 0$ is uniform), for the eliminative and guess-test learners at $C = 10$, by target selection. Compare the fast-mapping analysis in Figure 2.

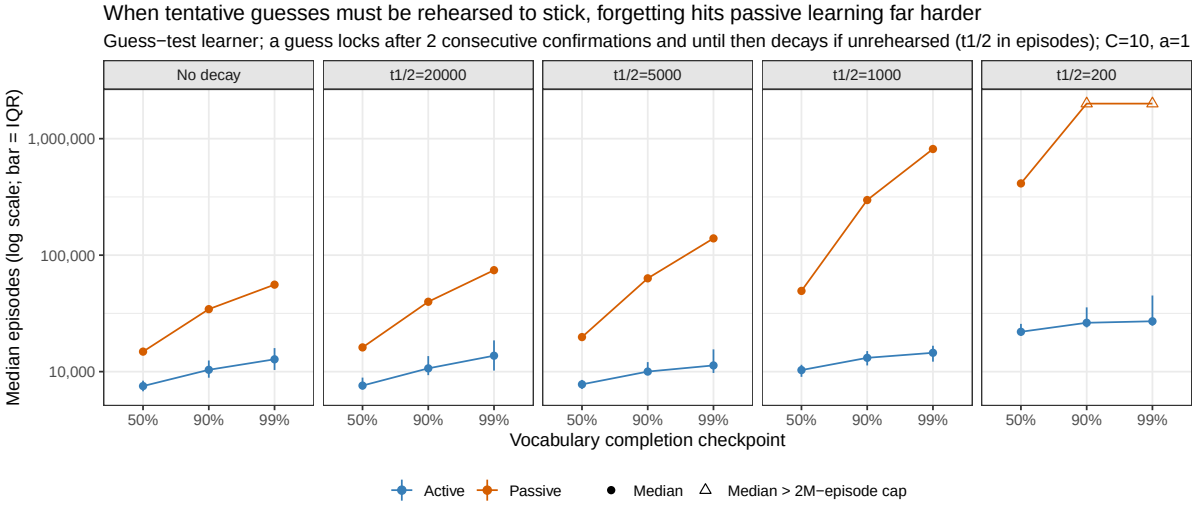


Figure 7. Median episodes (log scale; bars = interquartile range) for the guess-test learner to learn 50%, 90%, and 99% of the vocabulary, active vs. passive target selection, $C = 10$, $a = 1$, without forgetting and with half-lives of 20,000, 5,000, 1,000, and 200 episodes for guesses not yet locked in. Open triangles mark cells in which the median run did not reach the checkpoint within two million episodes.

Appendix A

Learning time for an arbitrary word frequency distribution

Suppose each episode presents word i with probability p_i, independently across episodes. Word i has occurred at least once in T episodes with probability $1-(1-p_i)^T$. Treating these events as independent across words gives Equation 2, whose left-hand side approximates the probability that every word has occurred. The events are in fact negatively associated, so the product slightly overstates this probability and the resulting T slightly understates the true value; the approximation becomes exact when the number of episodes is Poisson-distributed (with $e^{-p_i T}$ in place of $(1-p_i)^T$), and is accurate for the large vocabularies considered here.

For uniform frequencies the root is $T = f_{W,\epsilon}(1/W)$. For general p, the left-hand side of Equation 2 is increasing in each p_i, so replacing every p_i by $\min p$ or by $\max p$ brackets the root, giving Equation 4.

For a Zipf distribution with exponent a, $\min p = W^{-a}/H_{W,a}$, and

$$\frac{\partial \log T_+}{\partial a} = \frac{p_{\min}\left(\frac{\partial H_{W,a}}{\partial a}/H_{W,a} + \log W\right)}{(1-p_{\min})\log(1-p_{\min})}, \qquad \frac{\partial^2 \log T_+}{\partial a^2} \geq 0,$$

so $\log T_+$ is convex in a and T_+ grows faster than exponentially with the exponent.

Appendix B

Optimal scene choice by iterated linear programming

The idea is to raise the weakest word-referent pair step by step. Start from any scene distribution q_0 and find the pair k_1 least likely under it. Among all q that do not make k_1 less likely, choose the one that makes k_1 most likely; call it q_1. Then find the next-weakest pair k_2 under q_1, add the constraint that neither k_1 nor k_2 may get worse, and re-optimize. Each step guarantees one more pair without loosening earlier guarantees, so after at most W steps no pair can be improved without worsening another that is already at least as unlikely, which is the max-min condition defining $\hat{q}$.

Formally, write P_i for the ith row of P and $I = \{1, \ldots, W\}$. Set $K_0 := \emptyset$, $C_0 := \mathbb{S}^S$, $q_0 := e_1$. For $0 < n \leq W$, let

$$\begin{aligned} k_n &:= \operatorname*{arg\,min}_{k \in I \setminus K_{n-1}} P_k q_{n-1}, \quad K_n := K_{n-1} \cup \{k_n\}, \\ C_n &:= \{q \in C_0 \mid (P_{k_n} - P_k)q \leq 0 \text{ for all } k \in K_n\}, \\ q_n &:= \operatorname*{arg\,max}_{q \in C_n} P_{k_n} q, \end{aligned}$$

stopping at the first m with $\min_{k \in K_m} P_k q_m \leq \min_{k \in I} P_k q_m$, and output $\hat{q} := q_m$.

Random scene preferences. Because $\min(Pq)$ is concave in q, Jensen's inequality gives $\int_{q \in \mathbb{S}^S} \min(Pq)(S-1)!\, \mathrm{d}q \leq \min(P\mathbf{1}_S/S)$: a learner whose scene probabilities are drawn uniformly from the simplex does no better on average than one who chooses scenes uniformly.

Online estimation. The online learner of the main text applies the same algorithm to a sample estimate $\hat{P}$ of P. A smooth alternative replaces $\min(Pq)$ by the soft minimum $g_\lambda(q) = \sum_i P_i q\, e^{-\lambda P_i q} / \sum_i e^{-\lambda P_i q}$, which approaches $\min(Pq)$ as $\lambda \to \infty$ and whose gradient approaches $P_{\min}^T$, the row of the currently rarest pair. Gradient ascent on g_λ therefore shifts scene probability toward scenes in which the currently rarest pair is common, which is why estimates from a growing sample move q in the right direction, although this procedure is not guaranteed to converge exactly to $\hat{q}$.